CHAPTER 4:

Heat Death

VALENTINE: The heat goes into the mix.

(He gestures to indicate the air in the room, in the universe.)

THOMASINA: Yes, we must hurry if we are going to dance.

Tom Stoppard, *Arcadia*

One of my earliest astronomy memories is of a 1995 *Discover Magazine* cover story proclaiming a “Crisis in the Cosmos.” Something impossible was showing up in the data: the universe appeared to be younger than some of its own stars.

All the careful calculations of the age of the universe, based on extrapolating the current expansion back to the Big Bang, suggested the universe was somewhere in the vicinity of 10 or 12 billion years old, whereas measurements of the oldest stars in nearby ancient clusters gave a number closer to 15. Of course, estimating the ages of stars is not always an exact science, so there was a chance that better data might show that the stars were a bit younger than they looked, shaving maybe a billion or two years off the discrepancy. But extending the age of the universe to finish solving that problem would create an even bigger one. Making the universe older would have required scrapping the theory of cosmic inflation—one of the most important breakthroughs in the study of the early universe since the discovery of the Big Bang itself.

It would take another three years of combing through data, revising theories, and creating entirely new ways of measuring the cosmos before astronomers would find a solution that didn’t break the early universe. It just broke everything else. In the end, the answer came down to a new kind of physics woven into the very fabric of the cosmos—one that would fundamentally change our view of the universe and completely rewrite its future.